

H2 Chemistry

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Preface

This is my attempt to summarise most of the definitions from our entire H2 JC1 Chemistry syllabus into one document. I have not studied our school's lecture notes at all, so my definitions may be different from yours. If you find an error in this manuscript, please email (or text me if you have my number).

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I made these in a rush, so I have left out some of the less important material from this compilation. If you want to share the link to these notes with friends from another school, please ask for my permission.

Kieron

Best of luck for your Promos!

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1 Atomic Structure

Isotopes

Isotopes are atoms of the same element that have the *same atomic number* (that is, the same number of protons), but *different* mass numbers.

We say that two species are *isoelectronic* if they possess the same number of electrons.

Example:

For instance, consider the H_2O molecule. Here, oxygen has 8 electrons ($1s^2 2s^2 2p^4$), and each hydrogen carries one electron. Consequently, H_2O has 10 electrons.

Now, the electronic configuration of fluorine is $1s^2 2s^2 2p^5$ (i.e. F has 9 electrons), so F^- will have 10 electrons.

H_2O and F^- are, therefore, isoelectronic.

Similarly, say that two species are *isotonic* if they contain the same number of neutrons, and *isotopic* if they have the same number of protons.

Ionisation Energy

The *first ionisation energy* of an element is defined as the amount of energy required to remove one electron from each atom in a mole of gaseous atoms, producing one mole of gaseous cations.

For example, the first ionisation energy of Na can be represented through the equation



Ionisation energies are *always* positive because energy is required to remove an electron from an atom. This is because electrons in an atom are attracted to the positively charged nucleus.

2 Chemical Bonding

2.1 Types of Bonds

Definition: Ionic Bonds

An *ionic bond* is the electrostatic force of attraction between two oppositely charged ions formed as a result of the complete transfer of one or more electrons from one species to another.

For example, the ionic bond in NaCl is the electrostatic attraction between the positively charged Na^+ cations and the negatively charged Cl^- anions.

Definition: Covalent Bonds

A *covalent bond* between two atoms is the electrostatic attraction between the two positively charged nuclei and a shared pair of electrons between them.

If *both* the electrons in a covalent bond come from *only one* of the atoms involved in its formation, the bond is referred to as a *dative covalent* bond. Apart from the way it is formed, a dative bond is identical to an ordinary covalent bond¹.

Definition: Metallic Bonds

A *metallic bond* is the electrostatic force of attraction that two neighbouring metal ions (cations) have for the delocalised electrons between them.

Definition: Electronegativity

The *electronegativity* of an atom is a measure of its ability to attract a pair of shared electrons in a covalent bond towards itself.

¹Unfortunately, a satisfying explanation of this requires molecular orbital theory. That said, a covalent bond is formed by the attraction between the positively charged nuclei and the shared electron pair, so it shouldn't matter *where* the electron pair comes from.

Consequently, if the electronegativity *difference* between two atoms in a covalent bond is large, such as with Na and Cl, a covalent bond does not form because the majority of the electron density in the bond will be directed toward the species with higher electronegativity, thereby producing ionic character.

In any molecule, electron density need not be uniform, i.e., there may be uneven charge distribution on the surface of the molecule (this is due to quantum mechanics and the Schrödinger Equation). Hence, one side of the molecule can, at any moment, possess a partial positive charge, and the other, a partial negative charge. This is a dipole, which generates an induced electric field. This dipole can, therefore, induce another dipole in any neighbouring molecule (due to its electric field), which will result in the molecules being attracted to one another. This is how non-polar molecules attract one another, such as Br_2 , H_2 , and N_2 . These forces are referred to as instantaneous dipole-induced dipole forces. That is,

Instantaneous dipole-induced dipole forces are the weak electrostatic forces of attraction between molecules, which arise due to induced dipole attraction.

Definition: Hydrogen Bonds

Hydrogen bonds are electrostatic forces of attraction between an *electron-deficient* hydrogen atom bonded to a very electronegative atom (such as N, O, or F) and the lone pair of a neighbouring very electronegative atom (such as N, O, or F).

Hydrogen bonds are stronger than ordinary dipole-dipole attractions, but weaker than covalent and ionic bonds.

2.2 Bond Formation

Definition: σ Bonds

σ -bonds are formed when the orbitals of two atoms overlap *head-on*.

In a σ -bond, the electron density is concentrated between the two atomic nuclei due to the head-on overlap².

Definition: π Bonds

π -bonds are formed when the *p*-orbitals of two atoms overlap *sideways*.

²This is not entirely correct, and only holds for *bonding* orbitals. We leave the discussion of *antibonding* orbitals due to destructive overlap of the two wave functions ψ_1 and ψ_2 to the more advanced reader.

In a π -bond, the electron density is in the space alongside the two nuclei, separated by a nodal plane along the internuclear axis.

Hence, π -bonds are weaker than σ -bonds due to less effective orbital overlap (sideways for π , whereas σ -bonds are formed through head-on orbital overlap).

The full treatment of how molecules actually form bonds with one another is called *Molecular Orbital (MO) Theory*, and can be found in any undergraduate inorganic chemistry textbook.

2.3 Molecular Structure

The electric force between two like charges is repulsive. As such, electron pairs in a molecule (including bonding and lone pairs) will repel each other to move as far apart as possible³.

Definition: VSEPR Theory

VSEPR Theory is a way to predict the structure of a molecule and makes the following assumption:
lone pair-lone pair repulsion > lone pair-bond pair repulsion > bond pair-bond pair repulsion.

3 The Gaseous State

Assumptions of the Kinetic Theory of Gases

1. The gas consists of particles of negligible size (or volume) with respect to the vessel it is contained in.
2. The gas particles have negligible intermolecular forces between them.
3. The gas particles have negligible intermolecular forces of attraction.
4. Collisions between gas particles are perfectly *elastic*, which is to say that there is no loss in kinetic energy after the gas molecules collide.

Definition: Partial Pressure

Partial pressure is the pressure that a gas would exert if it alone occupied the container.

Dalton's Law

³This is a slight inaccuracy. What is actually minimised is the work done required to create the system.

Dalton's Law states that in a mixture of gases which do not chemically react, the total pressure is equal to the sum of the partial pressures of all the gases making up the mixture.

That is, if p_T denotes the total pressure of the gas, and we have a mixture of gases A, B, C, D, ..., then

$$p_T = p_A + p_B + p_C + p_D + \dots$$

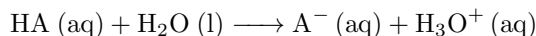
4 Acids and Bases

4.1 The Arrhenius Theory

Definition: Arrhenius Acids and Bases

An *Arrhenius acid* is a substance that dissociates in water to give hydrogen ions H^+ . An *Arrhenius base* is a substance that dissociates in water to give hydroxide ions OH^- .

Note that protons (naked H^+) cannot actually exist in aqueous solution as they are extremely unstable. Instead, they form the so-called hydronium ion (H_3O^+)⁴. For example,

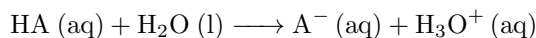


4.2 The Brønsted-Lowry Theory

Definition: Brønsted-Lowry Acids and Brønsted-Lowry Bases

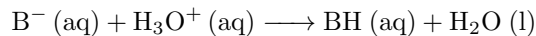
A *Brønsted-Lowry acid* is a species that donates a proton H^+ to a base; i.e., a proton donor. Conversely, a *Brønsted-Lowry base* is a species that accepts a proton from an acid; i.e., a proton acceptor.

Consider the dissociation of an acid HA:



We refer to the anion formed after proton transfer, A^- , as the *conjugate base*. Similarly, if B^- is a base,

⁴The hydronium ion can actually undergo further chemical reactions in aqueous solution, and the details can be found in an inorganic chemistry textbook.



BH is called the *conjugate acid* of the base B^{-} .

5 The Lewis Theory

Definition: Lewis Acids and Lewis Bases

A *Lewis acid* is an electron pair acceptor. On the other hand, a *Lewis base* is an electron pair donor.

You will realise that Lewis acids and Lewis bases are electrophiles and nucleophiles, respectively, after having learnt organic chemistry.

6 Stoichiometry

6.1 Fundamentals

Definition: Relative Atomic Mass (A_r)

The *relative atomic mass*, A_r , of an element is defined as the ratio of the average mass of one atom of the element to $\frac{1}{12}$ -th the mass of an atom of carbon-12.

Definition: Relative Isotopic Mass (A_r)

The *relative isotopic mass*, A_r , of an element is defined as the ratio of the average mass of one atom of the isotope to $\frac{1}{12}$ -th the mass of an atom of carbon-12.

Definition: Relative Molecular Mass (M_r)

The *relative molecular mass*, M_r , of a substance is defined as the ratio of the average mass of one molecule of the substance to $\frac{1}{12}$ -th the mass of an atom of carbon-12.

Definition: The Mole

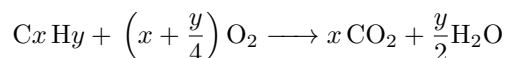
One *mole* of any substance is the amount of substance containing 6.02×10^{23} particles.

6.2 Compounds

Definition: Empirical and Molecular Formulae

The *empirical formula* of a compound is the *simplest formula* which shows the ratio of the atoms of the different elements in the compound. The *molecular formula* of a compound is the formula that shows the actual number of atoms of each element present in one molecule of the compound.

It is sometimes helpful to know the general relation for the combustion of a hydrocarbon. Let $x, y \in \mathbb{Z}^+$. Then,



7 Thermochemistry

An isolated system is a system that does not exchange matter or energy with its surroundings.

7.1 Enthalpy Changes

To measure enthalpy changes, we should have a set of conditions under which we measure them.

Standard Conditions refer to systems with the following.

1. Pressure of 1 bar (10^5Pa).
2. Temperature of 25°C .
3. Solution concentration of 1mol dm^{-3} .

Enthalpy Change of Formation

The *standard enthalpy change of formation*, ΔH_f° , of a compound is defined as the enthalpy change when *one mole* of the compound is formed from its elements under standard conditions.

If $\Delta H_f^\circ > 0$, then the compound is energetically less stable than its constituent elements (reactants), because energy *is needed* to form the compound.

Conversely, if $\Delta H_f^\circ < 0$, then the compound is energetically *more* stable than its constituent elements (reactants), because energy is *given out* to the surroundings.

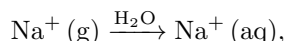
Definition: Enthalpy Change of Combustion (ΔH_c°)

The *standard enthalpy change of combustion*, ΔH_c° , is defined as the enthalpy change when *one mole* of a substance is *completely burnt* in oxygen under standard conditions (25°C, 10⁵Pa).

Definition: Enthalpy Change of Hydration ($\Delta H_{\text{hyd}}^\circ$)

The *standard enthalpy change of hydration*, $\Delta H_{\text{hyd}}^\circ$, of an ion is defined as the enthalpy change when one mole of gaseous ions is dissolved in a large amount of water under standard conditions (25°C, 10⁵Pa).

For example, in



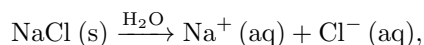
the standard enthalpy change of hydration is given by $\Delta H_{\text{hyd}}^\circ = -406 \text{ kJ mol}^{-1}$.

It should be noted that $\Delta H_{\text{hyd}}^\circ$ is *always negative*, as heat is evolved when bonds are formed between the ions and the water molecule dipoles.

Definition: Enthalpy Change of Solution ($\Delta H_{\text{sol}}^\circ$)

The *standard enthalpy change of solution*, $\Delta H_{\text{sol}}^\circ$, is defined as the enthalpy change when *one mole* of a substance dissolves in such a large volume of solvent that addition of more solvent produces no further heat change, under standard conditions (25°C, 10⁵Pa).

For example,



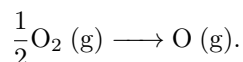
$$\Delta H_{\text{sol}}^\circ = +4.0 \text{ kJ mol}^{-1}.$$

Now, when $\Delta H_{\text{sol}}^\circ$ is very large (that is to say, very positive), it is energetically favourable for the compound to dissolve. Conversely, if $\Delta H_{\text{sol}}^\circ$ is negative, then it is energetically favourable for the compound to dissolve. **HOWEVER**, you should always calculate the Gibbs free energy to determine whether the dissolution process will be spontaneous.

Definition: Enthalpy Change of Atomisation ($\Delta H_{\text{at}}^\circ$)

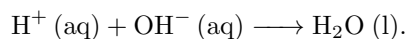
The *standard enthalpy change of atomisation*, $\Delta H_{\text{sol}}^\circ$, is defined as the enthalpy change when an element or a compound is converted into *one mole* of *free gaseous atoms* under standard conditions (25°C, 10⁵Pa).

This is illustrated in the reaction below, where $\Delta H_{\text{sol}}^\circ = +109\text{kJ mol}^{-1}$.

**Definition: Enthalpy Change of Neutralisation $\Delta H_{\text{n}}^\circ$**

The *standard enthalpy change of neutralisation*, $\Delta H_{\text{n}}^\circ$, is defined as the enthalpy change when *one mole* of water is formed in the neutralisation reaction between an acid and an alkali, the reaction being carried out in aqueous solution under standard conditions (25°C).

Note that for a neutralisation reaction between strong acids and strong bases, $\Delta H_{\text{n}}^\circ \approx -57.3\text{kJ mol}^{-1}$. This is because strong acids (such as HCl and HNO₃) and strong bases (such as NaOH and KOH) are completely ionised in aqueous solution, so the reaction we are concerned with is actually⁵



7.2 Energetics of Ionic Solids

Definition: Lattice Energy

The *lattice energy* of an ionic substance is the thermal energy evolved when *one mole* of the solid substance is formed from its separate gaseous ions under standard conditions.

All lattice energies are *negative* because thermal energy is released when the ions come together to form ionic bonds. For the people who take physics, recall that two opposite charges attract one another. Hence, the work done *by you* to bring them together is *negative*.

One should familiarise themselves with the formula

$$\text{LE} \propto \frac{q_+ q_-}{r_+ + r_-}.$$

⁵H⁺ in the reaction below should be the hydronium ion, but this may be ignored.

Again, if you take physics, this should be intuitive since the electric potential energy between two charges is

$$U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}.$$

For those interested, search up "Born-Landé Equation" to see how lattice energies can be approximated without experiment.

Definition: First Electron Affinity

The *first electron affinity* of an element is the enthalpy change when an electron is added to each atom in a mole of *gaseous* atoms to form singly-charged negative ions.

That is,



The second and third electron affinities are defined similarly.

Definition: Hess' Law

Hess' Law states that the enthalpy change in a chemical reaction is independent of the path or number of steps taken to achieve the reaction, provided that initial and final starting conditions remain the same.

In particular,

$$\Delta H = \sum_{k=1}^n \Delta H_k.$$

Definition: Entropy

Entropy, denoted by S^a , is a measure of a system's disorder.

^aWe denote entropy by this symbol in honor of the French engineer Sadi Carnot.

You can think of entropy as the number of ways to arrange things. For example, a system comprised of two coins will have a higher entropy than a system of one coin, because with two coins, we can have HH, HT, TT, whereas with one coin, we only have H or T. Here, H and T denote "heads" and "tails", respectively.

When the temperature of a chemical system increases, its entropy increases. With higher temperature, the Maxwell Boltzmann distribution (see 8.1) will

broaden, so the possible energies that particles can attain will increase, thereby increasing the system's disorder. Hence, entropy increases.

The more ways to arrange particles (or energy) in the system, the higher the system's entropy. Consequently,

$$S_{\text{gas}} \gg S_{\text{liquid}} > S_{\text{solid}}.$$

You will not be tested on this, but this is a useful law of nature.

The Second Law of Thermodynamics. *In an isolated system, entropy can only increase.*

In other words, the Second Law says that Nature prefers processes that increase a system's entropy.

8 Kinetics

Definition: Rate of Reaction

The *rate of reaction* is defined as the *change in concentration* of reactants or products per unit time.

Definition: Order of Reaction

The *order of reaction* with respect to a given reactant is defined as the power to which its concentration is raised to in the experimentally determined rate equation.

For example, in the rate equation

$$\text{rate} = k[\text{A}]^x[\text{B}]^y,$$

where A and B are reactants, the order of reaction with respect to A is x , and the order of reaction with respect to B is y .

The constant k in the equation above is referred to as the *rate constant*, and is constant at a particular temperature.

We say that a reaction is *zero order* with respect to a reactant if the rate of reaction is independent of the concentration of the reactant. In a *first order* reaction, the rate of reaction is proportional to *one* concentration term. The concentration-time graphs for a first order reaction have a constant half-life, determined by ⁶

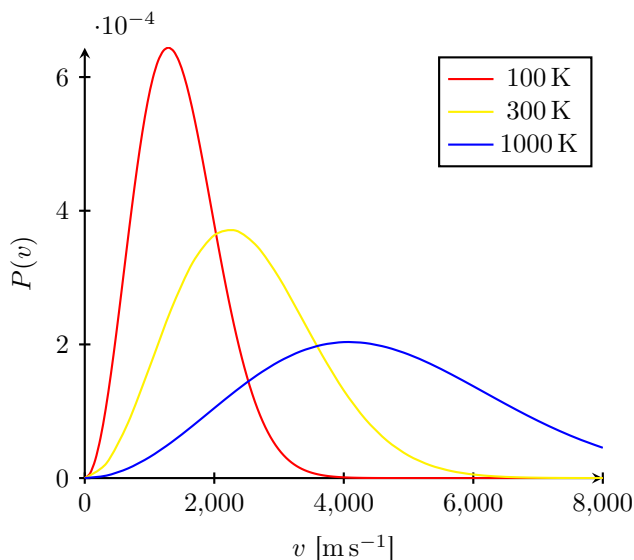
⁶For a proof, solve the differential equation $\frac{d[\text{A}]}{dt} = -k[\text{A}]$.

$$t_{1/2} = \frac{\ln 2}{k}.$$

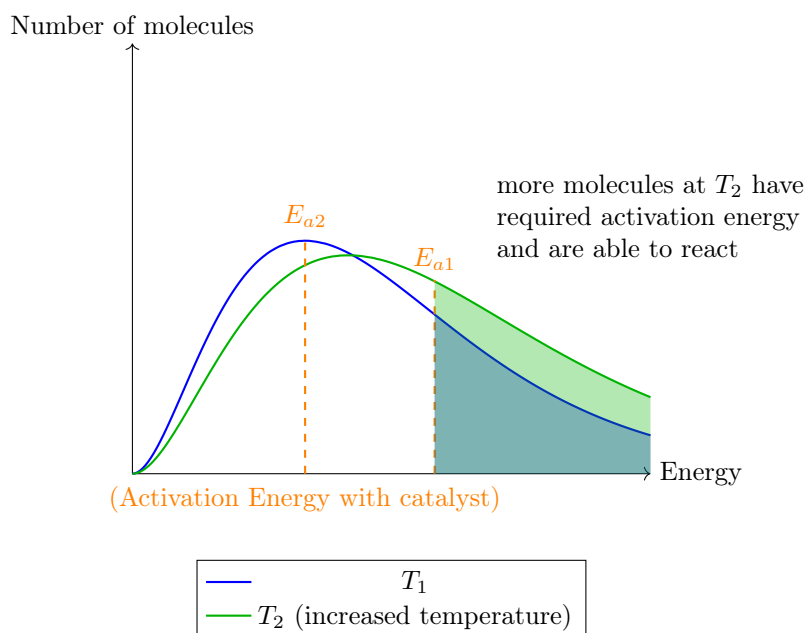
The half-life of a reaction is defined as the time taken for the initial concentration of a reactant to fall to *half* its value.

8.1 The Maxwell-Boltzmann Distribution

The graph below showcases the Maxwell-Boltzmann Distribution, which shows the fraction of particles that have a specific velocity. Now the *kinetic energy* of a particle (at non-relativistic speeds; i.e not moving extremely fast) is given by the formula $K = \frac{1}{2}mv^2$. This means the energy possessed by a reacting molecule is proportional to its speed squared. *The higher the speed, the higher the energy.* Now, according to *kinetic particle theory* that you should remember in Chapter 2 of the syllabus, the higher the temperature, the faster particles will move. This explains why a higher proportion of particles have higher speeds when the temperature is raised (see the graph below). Now recall our discussion on **The Collision Theory** back in Section 1. When more particles are of higher energy(due to increased temperature), the proportion of particles with energy greater than the *activation energy* E_a will increase. Therefore, the *rate of reaction* will increase.



A catalyst works on the same principle, except for the fact that a catalyst will *lower the activation energy* E_a , and will thus also increase the proportion of reacting molecules with energy which is greater than E_a . This is illustrated in the diagram below.



9 Equilibrium

Definition: Dynamic Equilibrium

In a reversible reaction, *dynamic equilibrium* occurs when the forward and reverse reactions are proceeding such that the rate of the forward reaction is equal to the rate of the reverse reaction.

Le Chatelier's Principle

Le Chatelier's Principle states that if a change is made to a system in equilibrium, the system will react in such a way as to tend to oppose the change, and a new equilibrium is formed.

Note that equilibrium constants are constant at a given temperature.

10 Useful Formulae and Identities

Ideal Gas Equation:

$$pV = nRT \quad (1)$$

Half-life for a first order chemical reaction:

$$t_{1/2} = \frac{\ln 2}{k}$$

where k denotes the rate constant.

Energy involved in the change in temperature ΔT of a substance of mass m with specific heat capacity x :

$$Q = mc\Delta T \quad (2)$$

Change in Gibbs Free energy:

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ \quad (3)$$

pH of a solution:

$$\text{pH} = -\log_{10}[\text{H}^+] \quad (4)$$

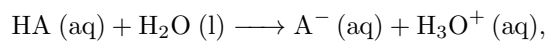
pOH of a solution:

$$\text{pOH} = -\log_{10}[\text{OH}^-] \quad (5)$$

At 25°C:

$$\text{pH} + \text{pOH} = 14 \quad (6)$$

For a weak acid HA, with



the acid dissociation constant is

$$K_a = \frac{[\text{A}^-][\text{H}_3\text{O}^+]}{[\text{HA}]}, \quad (7)$$

pK_a for an acid:

$$pK_a = -\log_{10} K_a. \quad (8)$$

and is constant at a given temperature.

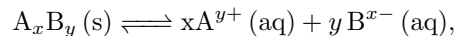
Point of maximum buffer capacity for an acidic buffer:

$$\text{pH} = pK_a.$$

Point of maximum buffer capacity in an alkaline buffer:

$$\text{pOH} = pK_b.$$

Solubility product for a sparingly soluble salt A_xB_y at equilibrium:

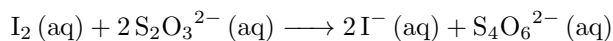
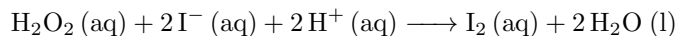


$$K_{sp} = [A^{y+}]^x [B^{x-}]^y. \quad (9)$$

This is constant at a particular temperature.

11 Additional Material

Iodine clock reaction (with starch):



I_2 produced by the first reaction reacts with the sodium thiosulfate, which is added to the solution. Once all thiosulfate ions are used up, an excess of iodine will form a starch-iodine complex, turning the solution deep blue.

Electron configurations of chromium and copper:

$$\begin{aligned} [Cr] &: 1s^2 2s^2 2p^2 3s^2 3p^6 3d^5 4s^1 \\ [Cu] &: 1s^2 2s^2 2p^2 3s^2 3p^6 3d^{10} 4s^1 \end{aligned}$$